

8(a)	$V^+ = 3.0 \times 3.0 / (2.5 + 3.0)$	C1
	$= 1.6 \text{ V}$	A1
8(b)	V^- is +2.0 V or $V^- > V^+$	B1
	output is negative so (LED) does not emit light	B1
8(c)	at 0 °C, $V^- = 1.7 \text{ V}$ or for all temperatures above 0 °C, resistance of thermistor < 4.2 kΩ	B1
	V^- always greater than V^+ (so no switching)	B1
8(d)	(at 20 °C,) $R_T = 1.8 \text{ k}\Omega$	C1
	$2.5 / 3.0 = 1.8 / R$ or $[R / (R + 1.8)] \times 3.0 = 1.6$	C1
	$R = 2.2 \text{ k}\Omega$	A1